

Retroarch3DS~RomMArch Front-End Intergration

All due credit to the developers of RetroArch and RomM. I take no credit for their respective masterworks.

This is a modified front-end of the latest Retroarch3DS Build, to intergrate synchronization with your RomM Database.

It is -only- a front-end replacement of RetroArch3DS. Retroarch3DS still installs cores independently of the front-end, within which the game plays and retains a separate configuration from the front-end.

Nothing changes with how your cores interact with this front-end.

A new sub-menu is now available only within the front-end, allowing synchronization with your RomM server.

Manual save synchronization is the supported save-sync mode in this release.

The **Automatic Synchronization** option is visible for developer testing only and is not functional with the currently released emulator CIA package. Leave it OFF/ignored for normal use.

Meaning: if I save a game on my 3DS and wish to push the new save to the RomM server, I MUST exit the game, re-open the retroarch front-end, navigate and execute the manual synchronization feature.

Likewise, if my local save is outdated, the manual synchronization feature must be executed before the new save is downloaded from RomM.

Use of this front-end assumes you have a working knowledge of: homebrew 3DS with FBI installer, and the retroarch client.

Menu Flow

```
RomMArch
├── RomM Configuration
│   ├── RomM Server Address
│   ├── RomM API Token
│   ├── HTTP Proxy: ON/OFF
│   └── Test Connection
├── Rom Synchronization
│   ├── Configure Local ROM Directory
│   └── RomM Emulator Library
└── Save Synchronization
    ├── Configure Core Save Directories
    │   └── Platforms [ON/OFF]...
    ├── [ ] Automatic Synchronization (non functional)
    └── Initiate Manual Sync
```

Preface

"RomMArch" uses its own configuration file located at /retroarch/rommarch.cfg, alongside your retroarch.cfg. RomMArch creates and maintains this file as configuration is entered.

After continued use, additional information will be written to rommarch.cfg, such as save hashes, which are critical to accurate save synchronization.

It should start out with something like:

```
server_url=http://192.168.1.195:####
```

```
api_token=rmm_123454321
```

```
roms_path=sdmc:/retroarch/downloads
```

Roms and Saves synchronize separately. Roms can and do download from RomM without synchronizing the RomM save. The reverse is untrue: saves cannot synchronize without their respective rom being downloaded/stored on the 3DS first.

Note the difference between Rom Synchronization and Save Synchronization, and their separate sub-menu's:

Rom Synchronization means the rom is on both your RomM server and your 3DS.

Save Synchronization reconciles save files for ROMs which are both on your 3DS AND in RomM. The ROM filenames must match exactly between the two locations.

ex: Chrono Trigger.smc and Chrono Trigger (USA).smc will not be recognized as the same ROM, nor synchronize cleanly.

Once you have the rom sync & save sync settings configured, use **Initiate Manual Sync** from the **Save Synchronization** menu to push/pull the save from your RomM database.

SAVE SYNCHRONIZATION CAN ONLY BE UTILIZED AFTER you've used your 3DS Emulator Core to launch at least one game.

The savefile pathway is generated the first time an emulator loads a rom. Some emulators will not populate a path until a game is actually saved once.

The default savefile path is: sdmc:/retroarch/cores/savefiles/core_name/savefile.srm

If you want to use this tool with an retroarch on your 3DS, but the emulator core hasn't established a savefile directory yet, GO DO THAT FIRST.

RomM Configuration

Server Address

Enter the RomM server base address. RomMArch accepts the server root and handles the API path internally. The value is persisted in rommarch.cfg.

API Token

Enter the RomM Client API token. The token is stored in rommarch.cfg and displayed verbatim in the configuration UI. The current project assumes a token with sufficient RomM access for ROM and save/device synchronization operations.

HTTP Proxy

This is useful for users who want Internet access from a mobile hotspot while on-the-go, particularly for **RetroAchievements** in compatible RomMARCH-integrated emulator builds.

Conduit is one third-party Android example that exposes a Wi-Fi Direct network and HTTP CONNECT proxy. It is not bundled with RomMARCH, and the projects are not affiliated with or endorsed by one another. Refer to the Conduit project for its own license and usage terms.

HTTP CONNECT carries TCP traffic; UDP-dependent networking is outside the scope of this feature.

Sleep/network recovery: after a prolonged sleep or network interruption, fully close and relaunch RomMARCH before retrying network-dependent functions. Waiting/retrying without restarting is not considered reliable.

HTTP Proxy is located inside **RomM Configuration**, between **RomM API Token** and **Test Connection**. Use the toggle to enable/disable proxy use; the proxy address and port are configured with the same RomMARCH setting.

RomMARCH stores the proxy values in `/retroarch/rommarch.cfg` as:

- **http_proxy_enabled**
- **http_proxy_host**
- **http_proxy_port**

The current default is **192.168.49.1:8080**. Always use the address and port reported by the proxy/hotspot software you are actually running.

Required setup:

1. Start the phone hotspot/proxy and connect the 3DS to it.
2. In the stock 3DS settings, open **System Settings** -> **Internet Settings** -> **Connection Settings**, select the active connection, choose **Change Settings**, move to the next page, and open **Proxy Settings**.
3. Enable the system proxy, enter the proxy address and port, save the connection, and run the stock 3DS **Connection Test**. The system test must succeed first.
4. Open RomMARCH, enable **HTTP Proxy**, enter the same address/port, then use **Test Connection**.
5. When returning to a normal Wi-Fi network that does not require a proxy, disable the RomMARCH proxy toggle (and use a non-proxy 3DS connection profile).

Test Connection

Test Connection performs an authenticated request to verify that the configured server and token work from the 3DS. This has been confirmed on real hardware.

When HTTP Proxy is enabled, Test Connection also verifies that RomMARCH can reach the configured RomM server through the active proxy path.

Rom Synchronization

Rom Library is gated by its prerequisites. Before the RomM emulator/rom browser is exposed, both a server address and API token must be setup in the former step, and a Local ROM Directory must be configured here.

The chosen directory is thereafter permanently stored in the `rommarch.cfg` file. Erasure of the value/config will gate the library access until reconfigured.

Local ROM Directory

default - `roms_path=sdmc:/retroarch/downloads`

This is the location where roms are stored on your 3DS. The directory can either be manually entered in your `cfg` file, or assigned through the interface. Erasure of the `.cfg` entry will gate access to your RomM Library until reconfigured.

RomM Emulator Library

Only visible after configuring Local Rom Directory.

Displays the emulator categories available on your RomM Server. Selecting an emulator opens that emulator's Rom Library.

Downloading/Synchronizing

After selecting an emulator, all available roms will be available for download or deletion to/from your local storage.

Roms are already considered synchronized if they already exist in both your local directory and your RomM Library, indicated with a [X] in the rom menu.

Unselecting an existing rom will prompt you to confirm deletion from local storage, with a double-check redundancy.

Deleting

Deleting a locally stored rom has no impact on your RomM Library. It only removes the local rom file from your configured directory. It also does NOT remove any save files that are already created locally.

To download a rom from your RomM Library, in the rom menu, select one/multiple, then choose Synchronize at the bottom of the page.

The game will download to the directory you configured. If you select a rom to download, but return to the Emulator directory without choosing Synchronize, then the selection will be cleared and it will not be downloaded.

It's faster the manually copy large files directly to your SD card directory, due to the limited processing speed.

If manually copying roms to your SD card rom library, be mindful of consistant file naming between RomM and 3DS storage, to ensure accurate synchronization.

For roms with multiple discs for which you wish to use an M3U configuration, see my note "Regarding M3U's" at the bottom.

Save Synchronization

Save FILES are synchronized.

Save states are explicitly NOT synchronized.

It was an intentional project boundary from the beginning, not an unfinished feature.

RomMArch synchronizes normal in-game save files only. RetroArch save states are not synchronized, modified, or deleted.

If your RomM save has a timestamp suffix, commonly something like "Chrono Trigger [timestamp information].sav", RomMArch will rename the save locally when it is pulled to your 3DS so that it is appropriately recognized by the emulator. The local save filename is derived from the ROM filename, while retaining the appropriate save extension.

Example: A time-stamped RomM save named "Chrono Trigger [timestamp information].sav" will be synchronized to the 3DS as "Chrono Trigger.sav".

PULL AND PUSH SAVE AUTHORITY ARE HANDLED DIFFERENTLY.

PULLING FROM RomM:

RomMArch uses RomM's server-side update information to determine the most recently updated save. The most recently updated eligible save on RomM is considered the newest remote save, regardless of its filename.

Extra words or timestamps written inside the filename do NOT determine which remote save is newest. RomM's server-side update information does.

For example, if RomM contains "Chrono Trigger.sav" and "Chrono Trigger [09-10-26 1818].sav", either file may be the newest remote save. The timestamp written in the second filename has no authority by itself. If RomM reports that file as the most recently updated save, RomMARCH can pull its contents and write them locally to the 3DS as the canonical "Chrono Trigger.sav".

PUSHING TO RomM:

Uploads from the 3DS are handled differently. RomMARCH pushes the local save to the canonical RomM save file whose base filename matches the ROM. For example: ROM "Chrono Trigger.smc" -> canonical save "Chrono Trigger.sav".

If RomM also contains timestamped or backup saves such as "Chrono Trigger [09-10-26 1818].sav" or "Chrono Trigger_previous.sav", those files may still be considered for a pull if RomM reports one of them as the newest remote save, but they are NOT used as normal upload destinations.

When the 3DS save is pushed to RomM, RomMARCH updates "Chrono Trigger.sav". Timestamped and backup copies are left untouched. If a canonical save does not yet exist on RomM, RomMARCH creates one rather than overwriting a timestamped or backup save.

In short:

PULL FROM RomM = newest eligible remote save wins.

PUSH TO RomM = canonical save filename is the destination.

The 3DS clock is NOT used to determine save authority.

Rooted 3DS systems commonly have an unconfigured or inaccurate date/time, and that's fine. The synchronization system is resilient to a wrong 3DS date/time.

Save sync only processes ROMs actually present on the 3DS.

Configure Core Save Directories

This tells RomM where to look for the selected console's save file.

NOTE: The core MUST be used to launch a game at least once. This GENERATES THE SAVEPATH, generally located in: `sdmc:/retroarch/cores/savefiles/core_name/savefile.srm`

You'll be required to point RomM to your emulator's save-path.

Due to various emulator cores producing different savepaths, and depending on the user, two people may have different save paths for the same console.

A list of your RomM Library Consoles will be displayed. You'll see the [OFF] status next to each by default. Complete configuration for synchronization will be indicated by [ON].

Choose which console saves you wish to synchronize. In that console's sub-menu, you'll see:

☐ Enable Save Sync

☐ Local Save Path: unconfigured

Complete the setup by enabling and setting the path:

☒ Enable Save Sync

☒ Local Save Path: `sdmc:/retroarch/cores/savefiles/<core-folder>/` && select "Choose this folder"

Automatic Synchronization (Developer Preview)

The **Automatic Synchronization** toggle is present in the frontend for developer testing only. It is not a supported end-user feature in this release and should remain **OFF**.

The currently released emulator CIA package does not yet contain the updated RomMARCH-integrated emulator builds required to ship the automatic per-ROM launch/close/replacement synchronization feature. It is intended to carry forward into a future release alongside the updated emulator CIA package.

Initiate Manual Sync

With:

- RomM server and api configured
- a rom synchronized on both RomM and the 3DS
- your save sync enabled & pathway setting

.. RomMArch will scrub local and romM saves against each other and provide you with the results: # Pushed, # Pulled, # Remains the Same, # Conflict Detected

#Pushed = uploaded to RomM

#Pulled = downladed from RomM

#Remains the same = RomM and 3DS files match and nothing is transferred

#Conflict Detected = both the 3DS save and the selected RomM save have changed independently since their last known synchronized state, and their current contents differ.

You'll be given the choices:

choose the local save

choose the RomM save

do nothing

Notes

Regarding M3U's

This is **not** tested with m3u structure for multi-disc games. At the time of this build, I've experienced m3u save-file compatibility errors cross-platform, between the RomM interface and other devices. Therefore, my RomM database has all multi-disc games stored traditionally (Chrono Cross (Disc 1).bin) as separate rom/save directories, and they're stored likewise on my 3DS in the retroarch/downloads directory, without any m3u configuration. They speak to each other in this similar manner. I cannot guarantee that if you have M3U setup on your RomM and 3DS side, that they'll work together the same.

Traditional:

~decades of success

rom_name=Chrono Cross (Disc 1).bin

save_name=Chrono Cross (Disc 1).save_extension

M3U:

~ depending on your setup, a text file and/or folder named Chrono Cross.m3u, are configured with the individual discs, still named:

- Chrono Cross (Disc 1).bin
- Chrono Cross (Disc 2).bin

however, resulting in..

rom_name=Chrono Cross.m3u (regardless of disc)

save_name=Chrono Cross.save_extension (regardless of disc)

Without going much more in depth, again, I'm assuming users have a working knowledge regarding use of their own emulator to perform disc-swap functions, because the RomM interface doesn't play nice with it yet.

With all that said.. expect RomMArch to synchronize the save associated with the individual RomM ROM/disc entry.

RetroAchievements Menu Performance

On Nintendo 3DS, the Achievements menu no longer eagerly decodes badge textures that RGUI does not display. The first visit may still perform normal RetroAchievements/network initialization, but subsequent openings avoid the previous repeated badge-allocation cost and should respond substantially faster.

Quick Menu Reset

The Nintendo 3DS **Quick Menu** -> **Reset** path has been changed to resolve an indefinite frontend hang seen with static cores. Reset now reloads the current content inside the already-running integrated core instead of entering the problematic direct core-reset path.

This also avoids accidentally launching or installing a separate core CIA while restarting the current game.

Known Errors

HTTP-1

Closing the 3DS lid while the RomMArch frontend is running puts the console to sleep and interrupts its network connection. After reopening the lid, attempting to use RomMArch without restarting the app can produce the HTTP-1 error.

To recover, fully close and reopen the RomMArch frontend after the console wakes, then retry the operation. Reopening the lid alone does not restore the connection used by RomMArch.

To avoid this issue, close the frontend before putting the console to sleep, or restart it after waking before using RomMArch.